

Practical 1

Aim: Design Login activity and implement control events : Use EditText, Checkbox and Buttons

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout android:layout_margin="10dp"
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:orientation="vertical"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:textSize="25dp"
        android:text="Shree Swami Atmanand Saraswati Institute of Technology"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <ImageView
        android:id="@+id/imageView"
```

```
android:layout_width="match_parent"  
android:layout_height="271dp"  
app:srcCompat="@drawable/ssasitlogo" />
```

```
<EditText
```

```
    android:id="@+id/txt_u_name"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:ems="10"  
    android:inputType="text"  
    android:text="Enter Name" />
```

```
<EditText
```

```
    android:id="@+id/txt_pass"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:hint="Password"  
    android:ems="10"  
    android:inputType="textPassword" />
```

```
<CheckBox
```

```
    android:id="@+id/chk_rem"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Remember me" />
```

```

<Button
    android:id="@+id/btn_login"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login" />
</LinearLayout>

```

MainActivity.java:

```

package com.ssasit.pr1;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    EditText U_name;

    EditText Pass;

    CheckBox Remember;

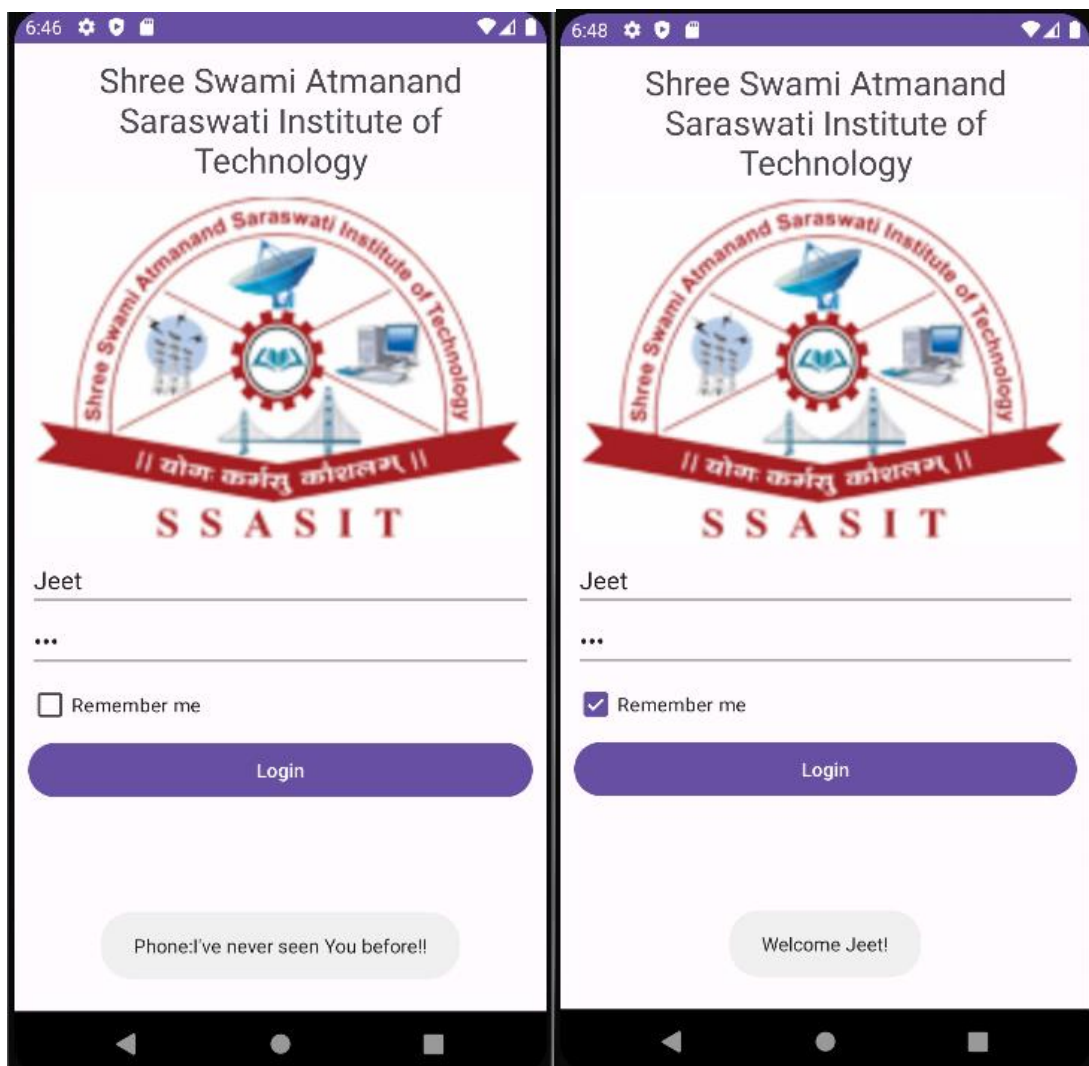
    Button login;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

```

```
U_name = findViewById(R.id.txt_u_name);
Pass = findViewById(R.id.txt_pass);
Remember = findViewById(R.id.chk_rem);
login = findViewById(R.id.btn_login);

login.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        if (Remember.isChecked()){
            Toast.makeText(MainActivity.this, "Welcome
"+U_name.getText().toString()+"!", Toast.LENGTH_SHORT).show();
        }else{
            Toast.makeText(MainActivity.this, "Phone:I've never seen You before!!",
            Toast.LENGTH_SHORT).show();
        }
    }
});
}
```

Output:

(without remember me button)

(with remember me button)

Practical 2

Aim: Implement Practical 1 using following layouts:

- 1. Linear Layout**
- 2. Relative Layout**
- 3. Table Layout**

Code:

1. Linear Layout:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout android:layout_margin="10dp"
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:orientation="vertical"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:textSize="25dp"
        android:text="Shree Swami Atmanand Saraswati Institute of Technology"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
```

```
app:layout_constraintStart_toStartOf="parent"  
app:layout_constraintTop_toTopOf="parent" />
```

```
<ImageView  
    android:id="@+id/imageView"  
    android:layout_width="match_parent"  
    android:layout_height="271dp"  
    app:srcCompat="@drawable/ssasitlogo" />
```

```
<EditText  
    android:id="@+id/txt_u_name"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:ems="10"  
    android:inputType="text"  
    android:text="Enter Name" />
```

```
<EditText  
    android:id="@+id/txt_pass"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:hint="Password"  
    android:ems="10"  
    android:inputType="textPassword" />
```

```
<CheckBox
```

```

    android:id="@+id/chk_rem"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Remember me" />

```

```

<Button
    android:id="@+id/btn_login"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login" />

```

```

</LinearLayout>

```

2. Relative Layout:

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="10dp"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"

```

```
android:layout_height="wrap_content"  
android:gravity="center"  
android:textSize="25dp"  
android:text="Shree Swami Atmanand Saraswati Institute of Technology"  
android:layout_centerHorizontal="true"  
android:layout_alignParentTop="true" />
```

```
<ImageView
```

```
    android:id="@+id/imageView"  
    android:layout_width="match_parent"  
    android:layout_height="271dp"  
    android:layout_below="@id/textView"  
    android:layout_marginTop="10dp"  
    app:srcCompat="@drawable/ssasitlogo" />
```

```
<EditText
```

```
    android:id="@+id/txt_u_name"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:layout_below="@id/imageView"  
    android:layout_marginTop="10dp"  
    android:ems="10"  
    android:inputType="text"  
    android:text="Enter Name" />
```

```
<EditText
```

```
android:id="@+id/txt_pass"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:layout_below="@id/txt_u_name"  
android:layout_marginTop="10dp"  
android:hint="Password"  
android:ems="10"  
android:inputType="textPassword" />
```

<CheckBox

```
android:id="@+id/chk_rem"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:layout_below="@id/txt_pass"  
android:layout_marginTop="10dp"  
android:text="Remember me" />
```

<Button

```
android:id="@+id/btn_login"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:layout_below="@id/chk_rem"  
android:layout_marginTop="10dp"  
android:text="Login" />
```

</RelativeLayout>

3. Table Layout:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="10dp"
    tools:context=".MainActivity"
    android:stretchColumns="1">

    <TableRow>
        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center"
            android:textSize="25dp"
            android:text="Shree Swami Atmanand Saraswati Institute of
Technology"
            android:layout_span="2"/>
    </TableRow>

    <TableRow>
        <ImageView
            android:id="@+id/imageView"
            android:layout_width="match_parent"
```

```

        android:layout_height="271dp"
        android:layout_span="2"
        app:srcCompat="@drawable/ssasitlogo" />
</TableRow>
<TableRow>
    <EditText
        android:id="@+id/txt_u_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
        android:text="Enter Name"
        android:layout_span="2"/>
</TableRow>
<TableRow>
    <EditText
        android:id="@+id/txt_pass"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:ems="10"
        android:inputType="textPassword"
        android:layout_span="2"/>
</TableRow>

<TableRow>

```

```

<CheckBox
    android:id="@+id/chk_rem"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Remember me"
    android:layout_span="2"/>
</TableRow>

<TableRow>
    <Button
        android:id="@+id/btn_login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_span="2"/>
    </TableRow>
</TableLayout>

```

MainActivity.java:

```

package com.ssasit.pr1;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;

```

```

import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    EditText U_name;
    EditText Pass;
    CheckBox Remember;
    Button login;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        U_name = findViewById(R.id.txt_u_name);
        Pass = findViewById(R.id.txt_pass);
        Remember = findViewById(R.id.chk_rem);
        login = findViewById(R.id.btn_login);

        login.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                if (Remember.isChecked()){
                    Toast.makeText(MainActivity.this, "Welcome
"+U_name.getText().toString()+"!", Toast.LENGTH_SHORT).show();
                }else{
                    Toast.makeText(MainActivity.this, "Phone:I've never seen You
before!!", Toast.LENGTH_SHORT).show();
                }
            }
        }
    }
}

```

```

});
}
}

```

Output:



(without remember me button)



(with remember me button)

Practical 3

Aim: Create Activities & implement following

- 1. Implicit intent**
- 2. Explicit Intent**
- 3. StartActivityForResult**

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_margin="10dp"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="278dp"
        android:layout_height="wrap_content"
        android:layout_alignParentStart="true"
        android:layout_alignParentTop="true"
        android:layout_alignParentEnd="true"
        android:layout_alignParentBottom="true"
        android:layout_marginStart="67dp"
```

```
android:layout_marginEnd="67dp"  
android:layout_marginBottom="453dp"  
app:srcCompat="@drawable/ssasitlogo" />
```

<EditText

```
android:id="@+id/txturl"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:layout_alignParentTop="true"  
android:layout_alignParentBottom="true"  
android:layout_marginTop="10dp"  
android:layout_marginBottom="10dp"  
android:ems="10"  
android:hint="Enter URL"  
android:inputType="text" />
```

<Button

```
android:id="@+id/btnimp"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:layout_alignParentStart="true"  
android:layout_alignParentEnd="true"  
android:layout_alignParentBottom="true"  
android:layout_marginStart="113dp"  
android:layout_marginEnd="109dp"  
android:layout_marginBottom="254dp"
```

```
android:text="Implicit Intent" />
```

```
<Button
```

```
    android:id="@+id/btnexp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_alignParentStart="true"
    android:layout_alignParentEnd="true"
    android:layout_alignParentBottom="true"
    android:layout_marginStart="111dp"
    android:layout_marginEnd="111dp"
    android:layout_marginBottom="172dp"
    android:text="Explicit Intent" />
```

```
</RelativeLayout>
```

MainActivity.java:

```
package com.ssasit.pr3;

import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
```

```

public class MainActivity extends AppCompatActivity {

    EditText url;

    Button imp,exp;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        url = findViewById(R.id.txturl);
        imp = findViewById(R.id.btnimp);
        exp = findViewById(R.id.btnexp);

        exp.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent exintent = new Intent(MainActivity.this,SecondActivity.class);
                startActivity(exintent);
            }
        });

        imp.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent impintent = new Intent(Intent.ACTION_VIEW,
                Uri.parse("https://" + url.getText().toString()));

                startActivity(impintent);
            }
        });
    }
}

```

```

    }
    });
}
}

```

activity_second.xml:

```

<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="10dp"
    tools:context=".SecondActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentEnd="true"
        android:layout_alignParentBottom="true"
        android:layout_marginEnd="91dp"
        android:layout_marginBottom="373dp"
        android:text="Second Activity"
        android:textSize="80px" />

</RelativeLayout>

```

SecondActivity.java:

```

package com.ssasit.pr3;

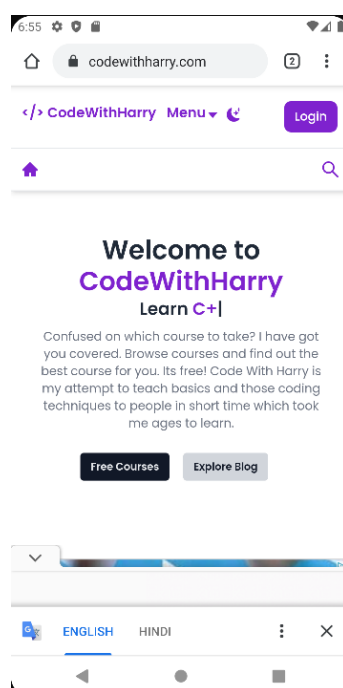
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}

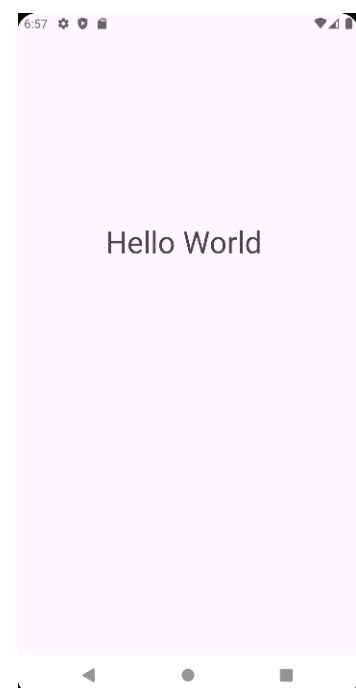
```

Output:

(Home page)



(Implicit Intent)



(Explicit Intent)

StartActivityForResult:**activity_main.xml:**

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_margin="10dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageView
        android:id="@+id/imageView2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:srcCompat="@drawable/ssasit_logo" />

    <TextView
        android:id="@+id/txtname"
        android:layout_width="107dp"
        android:layout_height="91dp"
        android:layout_marginTop="44dp"
        android:layout_weight="1"
        android:hint="Enter Message"
```

```

        app:layout_constraintBottom_toTopOf="@+id/btnsub"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.501"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/imageView2"
        app:layout_constraintVertical_bias="0.22" />
<Button
    android:id="@+id/btnsub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginBottom="208dp"
    android:layout_weight="1"
    android:text="Button"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.501"
    app:layout_constraintStart_toStartOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

MainActivity.java:

```

package com.ssasit.pr3_2;

import androidx.activity.result.contract.ActivityResultContracts;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;

```

```

import android.view.View;
import android.widget.Button;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity {
    TextView tv;
    Button btn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        tv = findViewById(R.id.txtname);
        btn = findViewById(R.id.btnsub);
        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(MainActivity.this,second_activity.class);
                startActivityForResult(intent,21);
            }
        });
    }
    @Override
    protected void onActivityResult(int requestCode, int resultCode, @Nullable
    Intent data) {
        super.onActivityResult(requestCode, resultCode, data);
        String msg = data.getStringExtra("msg");
        tv.setText(msg);
    }
}

```

```
}
```

activity_second.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="10dp"
    tools:context=".second_activity">

    <EditText
        android:id="@+id/etxtname"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
        android:text="Second Activity"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.272" />

    <Button
        android:id="@+id/btn"
        android:layout_width="133dp"
```

```

        android:layout_height="69dp"
        android:text="Button"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.501"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/etxtname"
        app:layout_constraintVertical_bias="0.166" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

SecondActivity.java:

```

package com.ssasit.pr3;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class SecondActivity extends AppCompatActivity

    @Override

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}

```

Output:

Practical 4

Aim: Implement activity Lifecycle and State Callbacks

Code:

MainActivity.java:

```
package com.ssasit.pr4;

import android.os.Bundle;
import android.util.Log;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private static final String tag = "LifeCycle";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        Log.d(tag, "onCreate");
        // Initialize your activity here
    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d(tag, "onStart");
    }

    @Override
    protected void onResume() {
        super.onResume();
        Log.d(tag, "onResume");
    }

    @Override
    protected void onPause() {
        super.onPause();
    }
}
```

```
    Log.d(tag, "onPause");
}
```

```
@Override
protected void onStop() {
    super.onStop();
    Log.d(tag, "onStop");
}
```

```
@Override
protected void onDestroy() {
    super.onDestroy();
    Log.d(tag, "onDestroy");
}
```

```
@Override
protected void onRestart() {
    super.onRestart();
    Log.d(tag, "onRestart");
}
}
```

Output:



```
2024-08-12 18:31:40.020 11770-11770 com.example.pr4 D Installing profile for com.example.pr4
2024-08-12 18:31:53.597 11770-11804 ProfileInstaller D onPause
2024-08-12 18:31:54.836 11770-11770 com.example.pr4 D onStop
2024-08-12 18:31:55.838 11770-11770 com.example.pr4 D onRestart
2024-08-12 18:32:32.574 11770-11770 com.example.pr4 D onStart
2024-08-12 18:32:32.577 11770-11770 com.example.pr4 D onResume
2024-08-12 18:32:32.579 11770-11770 com.example.pr4
```

Practical 5

Aim: Use an Options Menu

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="10dp"
    tools:context=".MainActivity">

    <com.google.android.material.appbar.MaterialToolbar
        android:id="@+id/mtool"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        app:layout_constraintBottom_toTopOf="@+id/imageView"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.0" />

    <ImageView
```

```

        android:id="@+id/imageView"
        android:layout_width="285dp"
        android:layout_height="271dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.504"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.145"
        app:srcCompat="@drawable/ssasit_logo" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

MainActivity.java:

```

package com.ssasit.pr5;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    Toolbar mn;

```

@Override

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    mn = findViewById(R.id.mtool);
    setSupportActionBar(mn);
}
```

@Override

```
public boolean onCreateOptionsMenu(Menu menu) {
    MenuInflater mnu = getMenuInflater();
    mnu.inflate(R.menu.menu_item, menu);
    return super.onCreateOptionsMenu(menu);
}
```

@Override

```
public boolean onOptionsItemSelected(@NonNull MenuItem item) {
    int id = item.getItemId();
    if (id == R.id.profile){
        Toast.makeText(this, "Profile", Toast.LENGTH_SHORT).show();
        return true;
    } else if (id == R.id.info) {
        Toast.makeText(this, "info", Toast.LENGTH_SHORT).show();
        return true;
    } else if (id == R.id.setting) {
        Toast.makeText(this, "setting", Toast.LENGTH_SHORT).show();
    }
}
```

```

        return true;
    } else if (id == R.id.aboutus) {
        Toast.makeText(this, "About US", Toast.LENGTH_SHORT).show();
        return true;
    } else {
        Toast.makeText(this, "Logout", Toast.LENGTH_SHORT).show();
        return super.onOptionsItemSelected(item);
    }
}
}
}

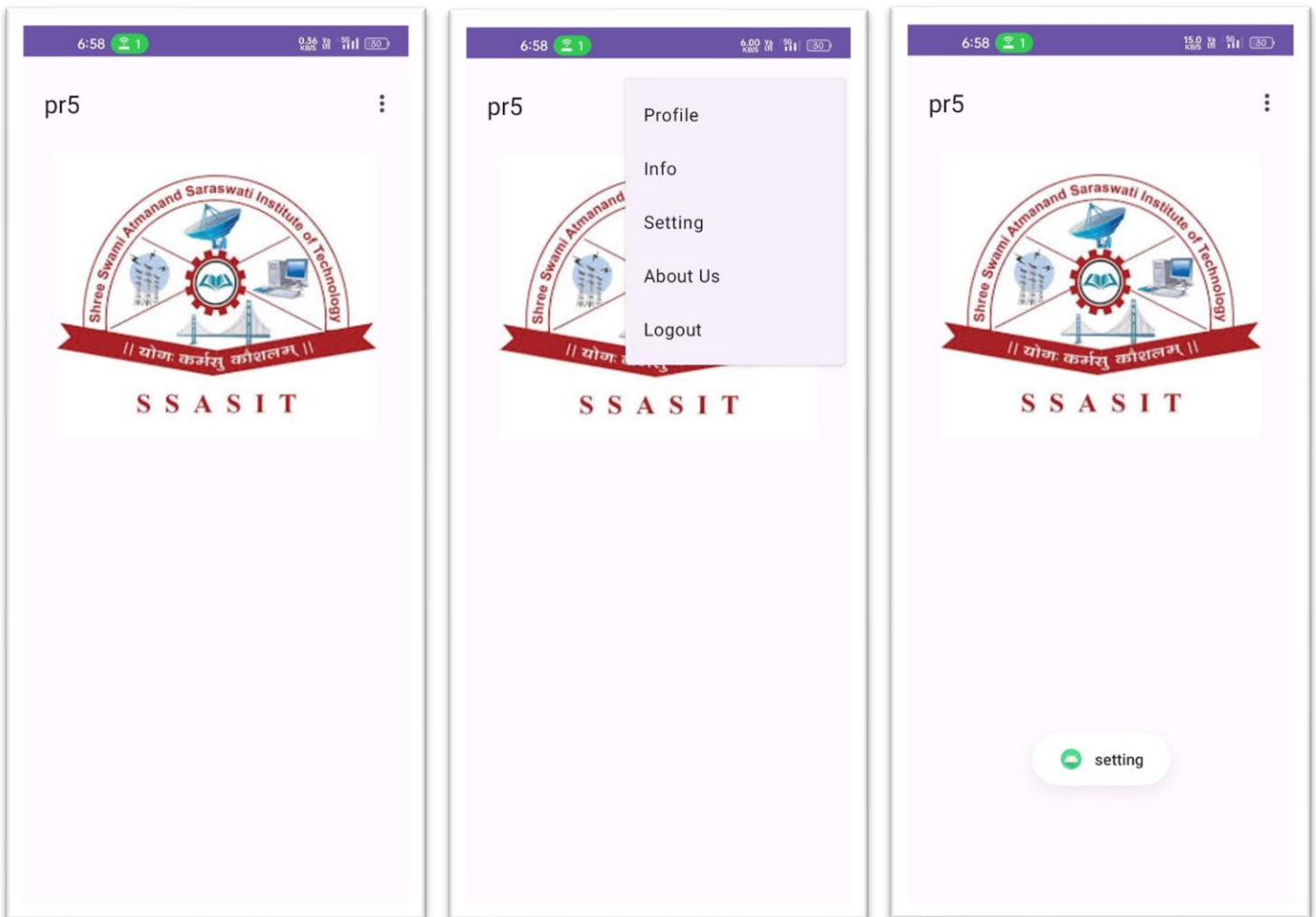
```

Menu_item.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:title="Profile"
        android:id="@+id/profile"/>
    <item android:title="Info"
        android:id="@+id/info"/>
    <item android:title="Setting"
        android:id="@+id/setting"/>
    <item android:title="About Us"
        android:id="@+id/aboutus"/>
    <item android:title="Logout"
        android:id="@+id/logout"/>
</menu>

```

Output:

Practical 6

Aim: Create a Recycler View and list the details of student using following fields: 1. Name 2. Address 3. Photo (Image) 4. Delete (Button Operation).

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_margin="10dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recycler_view"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

student_item.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_margin="10dp"
    android:orientation="horizontal"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    tools:context=".MainActivity">
```

```
<ImageView
    android:id="@+id/img"
    android:layout_width="50dp"
    android:layout_height="50dp"
    app:srcCompat="@drawable/sssit1" />
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:orientation="vertical">
    <TextView
        android:id="@+id/txtname"
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:text="Name"
        android:textSize="20dp" />
    <TextView
        android:id="@+id/txtaddress"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Address" />
</LinearLayout>
<Button
    android:id="@+id/btn"
    android:layout_width="wrap_content"
    android:layout_height="match_parent"
    android:text="Delete" />
</LinearLayout>
```

MainActivity.java:

```

package com.ssasit.pr6;

import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import android.os.Bundle;
import android.provider.CallLog;
import android.widget.Button;
import android.widget.TextView;
import java.util.ArrayList;
import java.util.List;

public class MainActivity extends AppCompatActivity {

    private RecyclerView recyclerView;
    private studentadapter adapter;
    private List<student> studentlist; // Here, <student> is datatype of list &
                                     // studentlist is object of list.

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        recyclerView = findViewById(R.id.recycler_view);
        recyclerView.setLayoutManager(new LinearLayoutManager(this));
        studentlist = new ArrayList<>(); // Generate Empty array list, to use during runtime.
        studentlist.add(new student("Jeet","123, ABC Street, Surat",R.drawable.img1));
        studentlist.add(new student("WIR","245, BBC Street, Delhi",R.drawable.img2));
        studentlist.add(new student("GKD","765, HDS Street, Pune",R.drawable.img1));
        studentlist.add(new student("ABC","356, FSE Street, Rajkot",R.drawable.img2));
        studentlist.add(new student("PQR","687, YRE Street, Gandhinagar",R.drawable.img1));
        studentlist.add(new student("XYZ","456, HFD Street, Baroda",R.drawable.img2));
    }
}

```

```

        studentlist.add(new student("LMA","344, HKG Street, Rajkot",R.drawable.img1));
        studentlist.add(new student("SDK","234, LTK Street, Mumbai",R.drawable.img2));
        studentlist.add(new student("KFD","133, PRE Street, Mumbai",R.drawable.img1));
        studentlist.add(new student("AKR","456, HIT Street, Rajkot",R.drawable.img2));
        adapter = new studentadapter(this,studentlist,this::onDeleteClick);
        recyclerView.setAdapter(adapter);
    }

    public void onDeleteClick(int position){
        studentlist.remove(position);
        adapter.notifyItemRemoved(position);
    }
}

```

student.java:

```

package com.ssasit.pr6;

public class student {
    String name,address;
    int image;

    public student(String name, String address, int image) {
        this.name = name;
        this.address = address;
        this.image = image;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }
}

```

```

public String getAddress() {
    return address;
}

public void setAddress(String address) {
    this.address = address;
}

public int getImage() {
    return image;
}

public void setImage(int image) {
    this.image = image;
}
}

```

studentadapter.java:

```

package com.ssasit.pr6;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.appcompat.view.menu.MenuView;
import androidx.recyclerview.widget.RecyclerView;
import java.util.List;

```

```

public class studentadapter extends
RecyclerView.Adapter<studentadapter.studentviewholder> {

    private Context context;

    private List<student> studentlist;

    private onDeleteClickListener onDeleteClickListener;

    public studentadapter(Context context, List<student> studentlist, onDeleteClickListener
onDeleteClickListener) {

        this.context = context;

        this.studentlist = studentlist;

        this.onDeleteClickListener = onDeleteClickListener;

    }

    @NonNull

    @Override

    public studentviewholder onCreateViewHolder(@NonNull ViewGroup viewGroup, int i) {
// Here, viewGroup is parent.

        View view =
LayoutInflater.from(context).inflate(R.layout.student_item,viewGroup,false); //pass the
parent here.

        return new studentviewholder(view,onDeleteClickListener);

    }

    @Override

    public void onBindViewHolder(@NonNull studentviewholder studentviewholder, int i) {
//Here, int i is the position.

        student student = studentlist.get(i); // pass the position here.

        studentviewholder.nametv.setText((student.getName()));

        studentviewholder.addresstv.setText((student.getAddress()));

        studentviewholder.photoid.setImageResource((student.getImage()));

    }

    @Override

    public int getItemCount() {

        return studentlist.size();

    }

```

```

    }

    public interface onDeleteClickListener{

        void onDeleteClick(int i); // Here, i is the position of item, which user wants to be
        deleted.

    }

    public class studentviewholder extends RecyclerView.ViewHolder{

        TextView nametv,addresstv;

        ImageView photoid;

        Button deletebutton;

        public studentviewholder(@NonNull View itemView, onDeleteClickListener
        onDeleteClickListener) {

            super(itemView);

            nametv = itemView.findViewById(R.id.txtname);

            addresstv = itemView.findViewById(R.id.txtaddress);

            photoid = itemView.findViewById(R.id.img);

            deletebutton = itemView.findViewById(R.id.btn);

            deletebutton.setOnClickListener(new View.OnClickListener() {

                @Override

                public void onClick(View view) {

                    if (onDeleteClickListener != null){

                        int position = getAdapterPosition();

                        if (position != RecyclerView.NO_POSITION){

                            onDeleteClickListener.onDeleteClick(position);

                        }

                    }

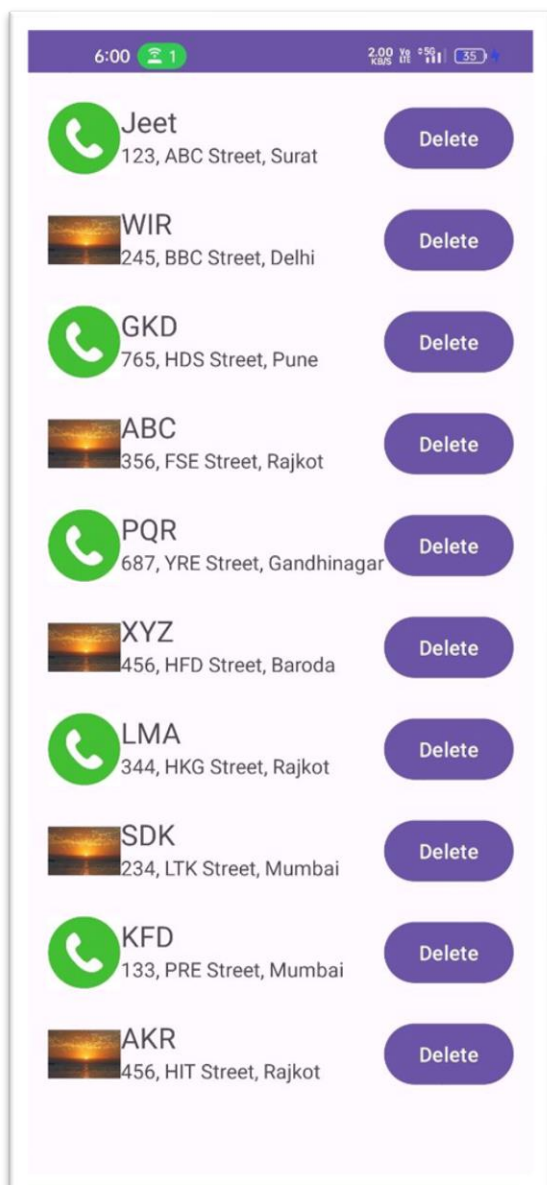
                }

            });

        }

    }
}

```

Output:**Initial Screen:****After Delete:**